Survival Analysis with R — Companion

STA310 · package and function guide by Collett chapter, with worked examples

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This document is an R companion to Collett (2023), organised chapter by chapter. For each topic it lists the functions used in the book (Chapter 17) together with more modern alternatives, and gives worked, runnable examples with interpretation using the book's own data sets. All packages are on CRAN; functions are written as `package::function` (base functions and S3 generics such as `summary()`, `plot()` and `coef()` are left unqualified). The `Surv()` term inside a model formula is also left unqualified — formula syntax requires the bare `Surv()` — whereas a `Surv` object created outside a formula is written `survival::Surv()`.

Chapters 1–16 are covered.

CRAN Packages used in the examples

Install once with:

```
install.packages(c(
  "broom",
  "glmnet",
  "gtsummary",
  "ggsurvfit",
  "survminer",
  "timeROC",
  "riskRegression",
  "rms",
  "flexsurv",
  "icenReg",
  "pseudo",
  "cmprsk"
))
```

The examples use the data sets that accompany the book (the folder "Data sets from Modelling Survival Data in Medical Research, 4th edition"). Download them here (click).

The relevant ones for Chapters 1–16 are:

- **IUD** (`iud`): time in weeks to discontinuation of an intrauterine device for 18 women (Example 1.1).
- **Multiple myeloma** (`myeloma`): survival of 48 patients, with seven covariates — age, sex, blood urea nitrogen (`bun`), serum calcium (`ca`), haemoglobin (`hb`), percentage of plasma cells (`pcells`) and `protein` (Example 1.3).
- **Breast cancer** (`breast`): survival of 45 women classified by HPA staining (`stain`) (Example 1.2).

```
data_dir <- "Data sets from Modelling Survival Data in Medical Research, 4th
edition"

iud <- read.table(
  file.path(data_dir, "Time to discontinuation of the use of an IUD.dat"),
  header = TRUE
)
myeloma <- read.table(
  file.path(data_dir, "Survival of multiple myeloma patients.dat"),
  header = TRUE
)
breast <- read.table(
```

```
  file.path(data_dir, "Prognosis for women with breast cancer.dat"),
  header = TRUE
)
breast$stain <- breast$stain - 1 # 0 = negative, 1 = positive HPA staining

# Additional data sets used in Chapters 5–16
hyper      <- read.table(file.path(data_dir, "Treatment of hypernephroma.dat"),
header = TRUE)                 # Ch. 6
ovarian     <- read.table(file.path(data_dir, "Chemotherapy in ovarian cancer
patients.dat"), header = TRUE)     # Ch. 7
bmt       <- read.table(file.path(data_dir, "Bone marrow transplantation.dat"),
header = TRUE)                 # Ch. 8
kidney         <- read.table(file.path(data_dir, "Survival following kidney
transplantation.dat"), header = TRUE)    # Ch. 10
gastric    <- read.table(file.path(data_dir, "Survival of patients with gastric
cancer.dat"), header = TRUE)     # Ch. 11
liver        <- read.table(file.path(data_dir, "Survival of liver transplant
recipients.dat"), header = TRUE)      # Ch. 12
liver_wait <- read.table(file.path(data_dir, "Time to death while waiting for a
liver transplant.dat"), header = TRUE) # Ch. 14
hep         <- read.table(file.path(data_dir, "Chronic active hepatitis.dat"),
header = TRUE)                    # Ch. 15
```

In all these files `status` is already coded 1 = event, 0 = right-censored.

Chapter 1 — Survival analysis

The `Surv` object and the survivor, hazard and cumulative hazard functions (§1.2, §1.4).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Create a survival object	`survival::Surv()`	—
Estimate $S(t)$ (Kaplan–Meier)	`survival::survfit()`	`prodlim::prodlim()`, `rms::npsurv()`
Estimate $H(t)$	`survival::survfit(type = "fleming-harrington")`	`prodlim::prodlim()`
Plot $S(t)$	`plot(survival::survfit(...))`	`ggsurvfit::ggsurvfit()`, `survminer::ggsurvplot()`
Smooth estimate of $h(t)$	—	`muhaz::muhaz()`, `bshazard::bshazard()`

The `Surv` object

A right-censored observation is stored as a pair (time, event indicator). `survival::Surv()` creates that object; the `+` marks censored times (here the first few women in the IUD study, Example 1.1):

```
survival::Surv(iud$time, iud$status)
```

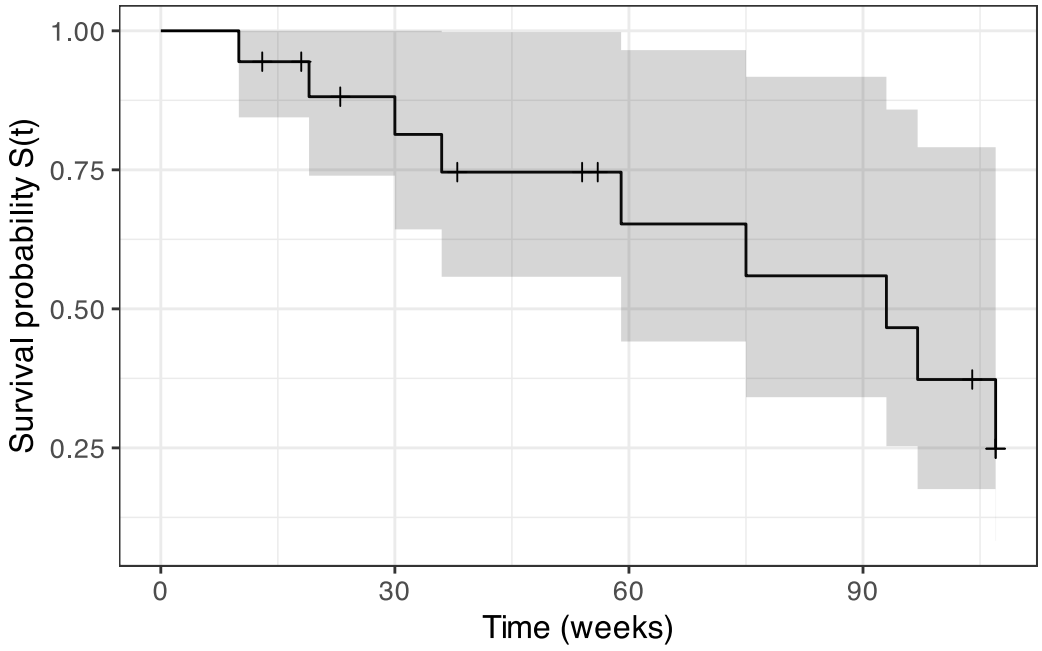
```
 [1]  10   13+  18+  19   23+  30   36   38+  54+  56+  59   75   93   97  104+
[16] 107  107+ 107+
```

The event indicator is `1` for an observed event and `0` for censoring, so the first four women contribute `10`, `13+`, `18+`, `19`: one discontinuation and two censored times before the next event.

Estimating and plotting $S(t)$ and $H(t)$

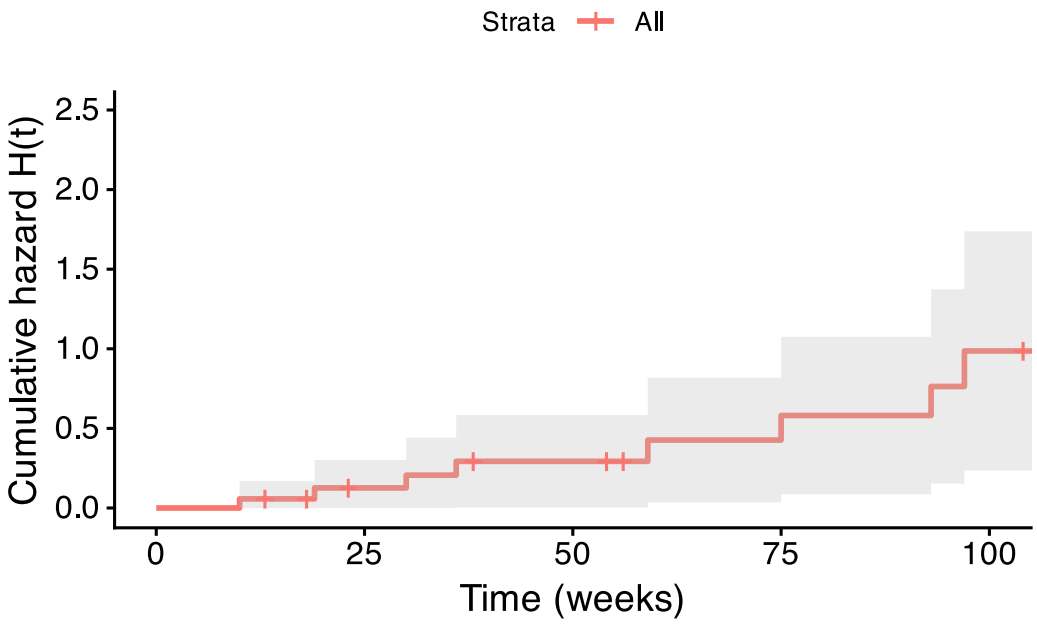
The Kaplan–Meier estimate of $S(t)$ is obtained from `survival::survfit()` and rendered with `ggsurvfit::ggsurvfit()`:

```
ggsurvfit::survfit2(Surv(time, status) ~ 1, data = iud) |>
  ggsurvfit::ggsurvfit() +
  ggsurvfit::add_confidence_interval() +
  ggsurvfit::add_censor_mark() +
  ggplot2::labs(x = "Time (weeks)", y = "Survival probability S(t)")
```



The cumulative hazard is plotted with `fun = "cumhaz"` (a `survminer::ggsurvplot()` transformation; `survival::survfit()` itself always returns survival):

```r
survminer::ggsurvplot(
  survival::survfit(Surv(time, status) ~ 1, data = iud),
  fun = "cumhaz",
  conf.int = TRUE,
  xlab = "Time (weeks)",
  ylab = "Cumulative hazard H(t)"
)
```



Interpretation. The survivor curve is a step function that drops only at observed discontinuation times and stays flat in between (censored observations reduce the risk set but do not cause a drop). The cumulative hazard is monotonically increasing; from (1.6) the two are linked by $S(t) = \exp\{-H(t)\}$. The estimated median survival time is read off where the curve crosses $S(t) = 0.5$.

Chapter 2 — Some non-parametric procedures

Kaplan–Meier and Nelson–Aalen estimation, medians, and group comparisons (§2.1–2.9).

Functions and packages

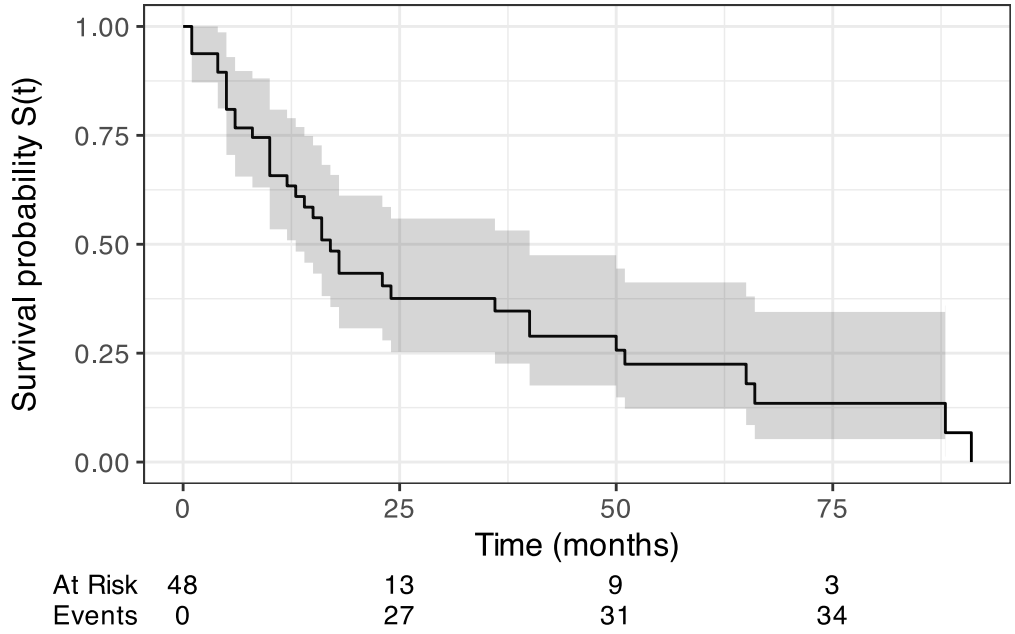
Task	Collett (Ch. 17)	Modern alternative
KM / Nelson–Aalen estimate	`survival::survfit(type = "kaplan-meier" / "fleming-harrington")`	`prodlim::prodlim()`, `rms::npsurv()`
SE and CI of $\hat{S}(t)$	`summary(survival::survfit(...))`	

Task	Collett (Ch. 17)	Modern alternative
Median and percentiles	`quantile(survival::survfit(...))`	
Two-group tests	`survival::survdiff()` (`rho = 0` log-rank, `rho = 1` Peto–Peto)	`coin::logrank_test()`, `FHtest::FHtestrcc()`
Plots with risk tables	`plot(survival::survfit(...))`	`ggsurvfit::ggsurvfit(+ add_risktable())`, `survminer::ggsurvplot()`

Kaplan–Meier estimate with risk table

We estimate the survivor function for the 48 multiple myeloma patients (Example 2.2):

```
ggsurvfit::survfit2(Surv(time, status) ~ 1, data = myeloma) |>
  ggsurvfit::ggsurvfit() +
  ggsurvfit::add_confidence_interval() +
  ggsurvfit::add_risktable() +
  ggplot2::labs(x = "Time (months)", y = "Survival probability S(t)")
```



The printed summary gives $\hat{S}(t)$, its standard error (Greenwood, eq. 2.12) and the confidence interval at chosen times:

```r
sfit2 <- survival::survfit(Surv(time, status) ~ 1, data = myeloma)
summary(sfit2, times = c(12, 24, 48))
```

```
Call: survfit(formula = Surv(time, status) ~ 1, data = myeloma)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   12     28      17    0.634  0.0710        0.509        0.789
   24     14      10    0.376  0.0762        0.252        0.559
   48      9       3    0.289  0.0732        0.176        0.475
```

```r
s24 <- summary(sfit2, times = 24)$surv
```

Interpretation. An estimated 38% of patients survive beyond 24 months. The risk table underneath the plot shows how the number at risk shrinks as patients die or are censored.

Nelson–Aalen and the median

The Nelson–Aalen estimator (eq. 2.5) is obtained with `type = "fleming-harrington"`; the estimated median and other percentiles come from `quantile()` (here for the IUD data, Example 2.8):

```r
med <- quantile(
  survival::survfit(Surv(time, status) ~ 1, data = iud),
  probs = c(0.25, 0.5, 0.75)
)
med
```

```
$quantile
 25  50  75
 36  93 107

$lower
25 50 75
19 59 97

$upper
 25  50  75
107  NA  NA
```

Interpretation. The estimated median time to discontinuation is 93 weeks, with lower 95% limit 59 weeks; the upper limit is undefined because the survival curve's upper confidence band never falls below 0.5 within the follow-up.

Comparing two groups

The log-rank test (`rho = 0`) and the Peto–Peto test (`rho = 1`) for the two breast-cancer staining groups (Examples 2.11–2.13):

```
lr <- survival::survdiff(Surv(time, status) ~ stain, data = breast, rho = 0)
peto <- survival::survdiff(Surv(time, status) ~ stain, data = breast, rho = 1)
lr
```

```
Call:
survival::survdiff(formula = Surv(time, status) ~ stain, data = breast,
    rho = 0)

         N Observed Expected (O-E)^2/E (O-E)^2/V
stain=0 13        5     9.57      2.18      3.51
stain=1 32       21    16.43      1.27      3.51

 Chisq= 3.5  on 1 degrees of freedom, p= 0.06
```

```
peto
```

```
Call:
survival::survdiff(formula = Surv(time, status) ~ stain, data = breast,
    rho = 1)

         N Observed Expected (O-E)^2/E (O-E)^2/V
stain=0 13      3.0      6.6      1.97      4.11
stain=1 32     15.7     12.1      1.08      4.11

 Chisq= 4.1  on 1 degrees of freedom, p= 0.04
```

Interpretation. The log-rank statistic is $\chi^2 = 3.5$ on 1 d.f., p = 0.06. The positively stained group has more observed deaths than expected under the null (21 vs 16.4), so positive staining is associated with worse survival, but the evidence is only modest and not significant at the 5% level. The Peto–Peto test, which down-weights the tail, gives $\chi^2 = 4.1$, p = 0.04 — a similar conclusion. The book's Wilcoxon test (weighting by n_j) is not in base `survival`; use `coin::logrank_test()` or `FHtest::FHtestrcc()` (both CRAN).

Chapter 3 — The Cox regression model

Fitting, inference, model checking and prediction (§3.1–3.14).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Fit a Cox model	`survival::coxph()`	`rms::cph()`, `glmnet::glmnet(family = "cox")`, tidymodels `censored`
HR, CI, p-values	`summary(survival::coxph(...))`	`broom::tidy()`, `gtsummary::tbl_regression()`
Baseline hazard / survival	`survival::basehaz()`	`survival::survfit(survival::coxph(...))`
PH assumption	`survival::cox.zph()`	`rms::cox.zph()`
Residuals	`residuals(..., type = "...")`	—
Lasso variable selection	`penalized::penalized()`	`glmnet::glmnet(family = "cox")`
Firth correction (separation)	—	`coxphf::coxphf()`
Concordance	`survival::concordance()`	`CPE::phcpe()` (Gönen–Heller), `pec::pec()`, `riskRegression::Score()`
Time-dependent ROC	`survivalROC::survivalROC()`	`timeROC::timeROC()`, `riskRegression::Score()`
Forest plot	—	`survminer::ggforest()`, `ggstats::ggcoef_model()`

Fitting a Cox model with a single binary covariate

Example 3.1 models the hazard of death for the breast-cancer patients in terms of HPA staining (0 = negative, 1 = positive):

```
fit_bc <- survival::coxph(Surv(time, status) ~ stain, data = breast)
summary(fit_bc)
```

```
Call:
survival::coxph(formula = Surv(time, status) ~ stain, data = breast)

  n= 45, number of events= 26

       coef exp(coef) se(coef)     z Pr(>|z|)
stain 0.9093    2.4827   0.5009 1.815   0.0695 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
      exp(coef) exp(-coef) lower .95 upper .95
stain     2.483     0.4028    0.9302     6.626

Concordance= 0.603  (se = 0.042 )
Likelihood ratio test= 3.88  on 1 df,   p=0.05
Wald test            = 3.3  on 1 df,   p=0.07
Score (logrank) test = 3.52  on 1 df,   p=0.06
```

```
hr_bc <- exp(coef(fit_bc))["stain"]
ci_bc <- exp(confint(fit_bc))["stain", ]
p_bc <- summary(fit_bc)$coefficients["stain", "Pr(>|z|)"]
```

Interpretation. The estimated hazard ratio for positive versus negative staining is 2.48 (95% CI 0.93–6.63, p = 0.07): the hazard of death at any time is about 2.48 times as high for a woman with a positively stained tumour, so positive staining indicates a poorer prognosis.

Fitting a Cox model with several covariates

Example 3.2 models the hazard of death for the myeloma patients in terms of all seven recorded covariates:

```
fit <- survival::coxph(
  Surv(time, status) ~ age + sex + bun + ca + hb + pcells + protein,
  data = myeloma
)
summary(fit)
```

```
Call:
survival::coxph(formula = Surv(time, status) ~ age + sex + bun +
    ca + hb + pcells + protein, data = myeloma)

  n= 48, number of events= 36

              coef exp(coef)  se(coef)      z Pr(>|z|)
age     -0.018056  0.982106  0.027833 -0.649 0.516521
sex     -0.249473  0.779211  0.403093 -0.619 0.535985
bun      0.022661  1.022919  0.006110  3.709 0.000208 ***
ca       0.013265  1.013353  0.132681  0.100 0.920363
hb      -0.133017  0.875450  0.068527 -1.941 0.052249 .
pcells  -0.001359  0.998642  0.006588 -0.206 0.836585
protein -0.683269  0.504964  0.429395 -1.591 0.111556
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
        exp(coef) exp(-coef) lower .95 upper .95
age        0.9821     1.0182    0.9300     1.037
sex        0.7792     1.2833    0.3536     1.717
bun        1.0229     0.9776    1.0107     1.035
ca         1.0134     0.9868    0.7813     1.314
hb         0.8755     1.1423    0.7654     1.001
pcells     0.9986     1.0014    0.9858     1.012
protein    0.5050     1.9803    0.2177     1.172

Concordance= 0.705  (se = 0.048 )
Likelihood ratio test= 17.53  on 7 df,   p=0.01
Wald test            = 20.01  on 7 df,   p=0.006
Score (logrank) test = 25.59  on 7 df,   p=6e-04
```

```
hr_bun <- exp(coef(fit))[["bun"]]
p_bun <- summary(fit)$coefficients["bun", "Pr(>|z|)"]
hr_hb <- exp(coef(fit))[["hb"]]
p_hb <- summary(fit)$coefficients["hb", "Pr(>|z|)"]
lt <- summary(fit)$logtest
```

Interpretation. Each coefficient is a log hazard ratio. The two covariates most clearly associated with survival are blood urea nitrogen and haemoglobin: a one-unit higher `bun` multiplies the hazard by 1.023 (p = 2^{-4}), while a one-unit higher `hb` multiplies it by 0.875 (p = 0.052). The other covariates (age, sex, serum calcium, plasma cells, protein) are not individually significant. The likelihood-ratio test (χ^2 = 17.5 on 7 d.f., p = 0.01) shows the model as a whole is useful.

Publication-ready tables

`broom::tidy()` turns the fit into a data frame; `gtsummary::tbl_regression()` produces a table ready for a report:

```
broom::tidy(fit, exponentiate = TRUE, conf.int = TRUE)
```

```
# A tibble: 7 × 7
  term    estimate std.error statistic  p.value conf.low conf.high
  <chr>      <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 age        0.982   0.0278    -0.649  0.517       0.930      1.04
2 sex        0.779   0.403     -0.619  0.536       0.354      1.72
3 bun        1.02    0.00611    3.71   0.000208    1.01       1.04
4 ca         1.01    0.133      0.1000 0.920       0.781      1.31
5 hb         0.875   0.0685    -1.94   0.0522      0.765      1.00
```

```
6 pcells     0.999   0.00659   -0.206  0.837       0.986      1.01
7 protein    0.505   0.429     -1.59   0.112       0.218      1.17
```

```
gtsummary::tbl_regression(fit, exponentiate = TRUE)
```

Characteristic	HR	95% CI	p-value
age	0.98	0.93, 1.04	0.5
sex	0.78	0.35, 1.72	0.5
bun	1.02	1.01, 1.04	<0.001
ca	1.01	0.78, 1.31	>0.9
hb	0.88	0.77, 1.00	0.052
pcells	1.00	0.99, 1.01	0.8
protein	0.50	0.22, 1.17	0.11

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio

Model checking: the proportional hazards assumption

```
cz <- survival::cox.zph(fit)
cz
```

```
         chisq df    p
age     0.1280  1 0.72
sex     0.7724  1 0.38
bun     0.8564  1 0.35
ca      0.3629  1 0.55
hb      0.0428  1 0.84
pcells  2.1903  1 0.14
protein 0.4325  1 0.51
GLOBAL  4.0974  7 0.77
```

Interpretation. `survival::cox.zph()` tests each covariate (and the model as a whole) for proportional hazards by correlating the Schoenfeld residuals with time. Here none of the covariates shows evidence against PH (the global p-value is 0.77), so the proportional-hazards assumption appears tenable for this model.

Concordance

The concordance (c-index) measures how often the model ranks a longer survivor above a shorter one:

```
cconc <- survival::concordance(fit)
cconc
```

```
Call:
concordance.coxph(object = fit)

n= 48
Concordance= 0.7049 se= 0.04816
concordant discordant     tied.x     tied.y    tied.xy
       614        257          0         20          0
```

Interpretation. A concordance of 0.705 means the model discriminates higher-risk from lower-risk patients reasonably well (0.5 is chance; values above about 0.7 are generally considered useful). The Gönen–Heller concordance probability estimate (`CPE::phcpe()`) is an alternative that does not depend on the censoring distribution. Time-dependent AUC and Brier scores are computed below with `timeROC::timeROC()` and `riskRegression::Score()`.

Time-dependent ROC and prediction error

`riskRegression::Score()` estimates the time-dependent AUC with bootstrap confidence intervals, comparing a risk model against the null (no-covariate) model:

```
sc <- riskRegression::Score(
  list(
    risk = survival::coxph(
      Surv(time, status) ~ bun + hb + age + sex,
      data = myeloma,
      x = TRUE
    )
  ),
  Surv(time, status) ~ 1,
  data = myeloma,
  times = c(12, 24, 36),
  metrics = "auc",
  plots = "ROC"
)
sc
```

```


Metric AUC:

Results by model:
```

```
Key: <model, times>
    model times    AUC  lower  upper
   <fctr> <num> <char> <char> <char>
1:   risk    12   74.7   59.1   90.3
2:   risk    24   58.5   40.0   76.9
3:   risk    36   61.4   42.8   80.1

Results of model comparisons:

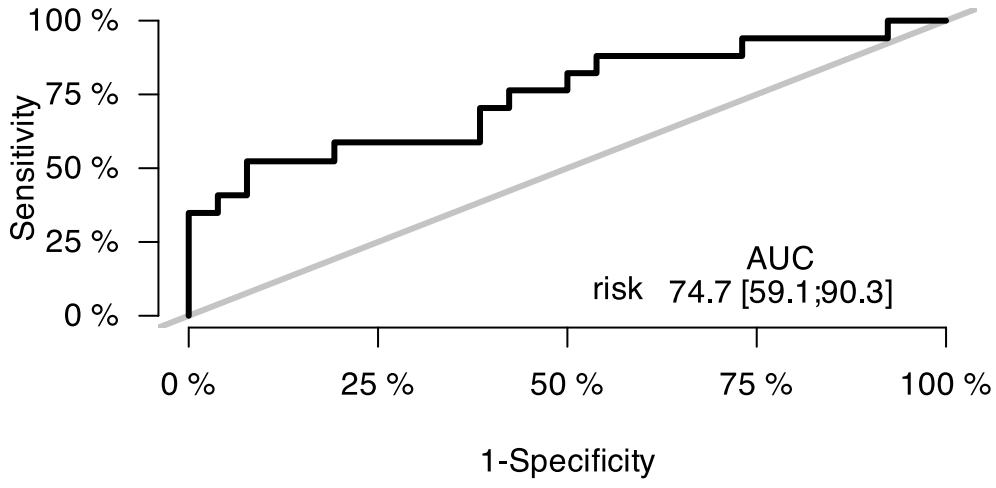
Empty        data.table        (0        rows        and        7        cols):
times,model,reference,delta.AUC,lower,upper...
```

```
Data for ROC plot are stored in the object as x[["ROC"]].
```

```
auc12 <- sc$AUC$score$AUC[sc$AUC$score$times == 12]
auc24 <- sc$AUC$score$AUC[sc$AUC$score$times == 24]
```

Interpretation. The AUC is about 74.7% at 12 months and 58.5% at 24 months, so discrimination is best in the short term. The ROC curves are drawn with `riskRegression::plotROC()`:

```
riskRegression::plotROC(sc, time = 12)
```



`timeROC::timeROC()` computes the same inverse-probability-of-censoring (IPCW) AUC directly from a single marker, without fitting a model. Note that haemoglobin is *protective* (higher = better survival), so it is negated to form a risk score (higher = worse); otherwise the AUC would fall below 0.5:

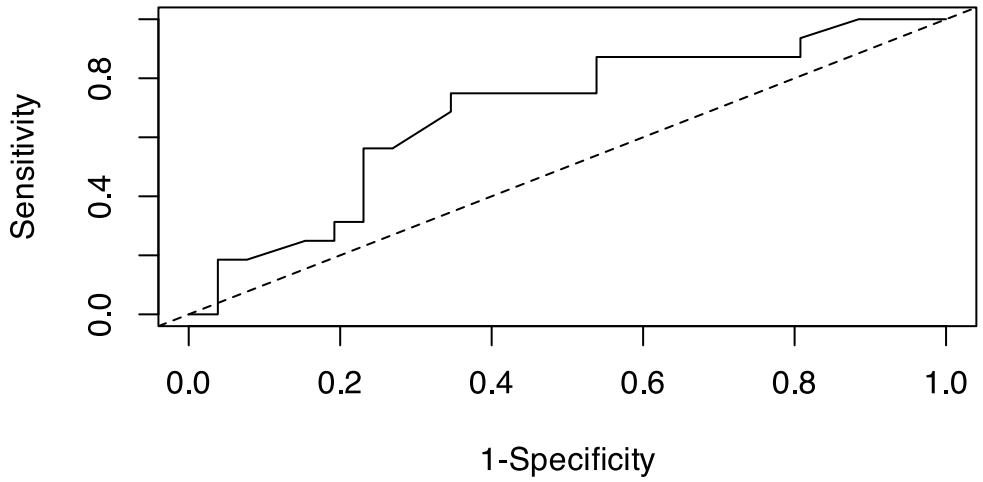
```
roc <- timeROC::timeROC(
  T = myeloma$time,
  delta = myeloma$status,
  marker = -myeloma$hb,
  cause = 1,
  weighting = "marginal",
  times = c(12, 24, 36),
  iid = TRUE
)
roc
```

```
Time-dependent-Roc curve estimated using IPCW  (n=48, without competing risks).
     Cases Survivors Censored AUC (%)    se
t=12    16        26        6   68.38  8.48
t=24    26        13        9   59.90  9.97
t=36    27        12        9   58.41 10.26

Method used for estimating IPCW:marginal

Total computation time : 0.13  secs.
```

```
plot(roc, time = 12, col = 1, title = "Time-dependent ROC at 12 months")
```



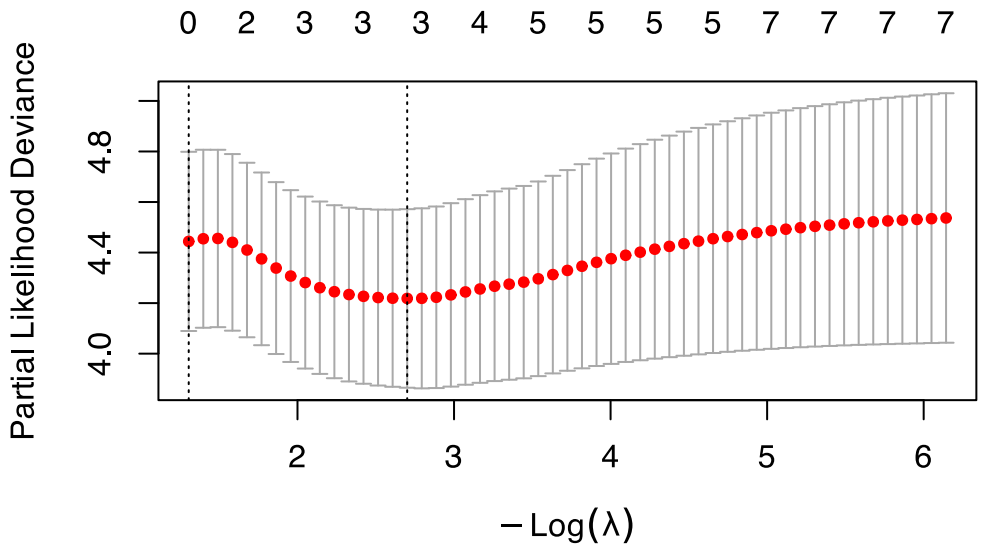
Interpretation. The marginal IPCW AUC based on haemoglobin alone is 68.4% at 12 months — weaker than the multivariate risk score, because a single covariate carries less information than the combined model.

Lasso variable selection

The lasso (eq. 3.16) shrinks coefficients and sets some to exactly zero. `glmnet::glmnet()` implements it for the Cox model; cross-validation selects λ (Example 3.7):

```
X <- as.matrix(myeloma[c("age", "sex", "bun", "ca", "hb", "pcells", "protein")])
Y <- survival::Surv(myeloma$time, myeloma$status)

cvfit <- glmnet::cv.glmnet(X, Y, family = "cox", cox.ties = "breslow")
plot(cvfit)
```




```
coef(cvfit, s = "lambda.min")
```

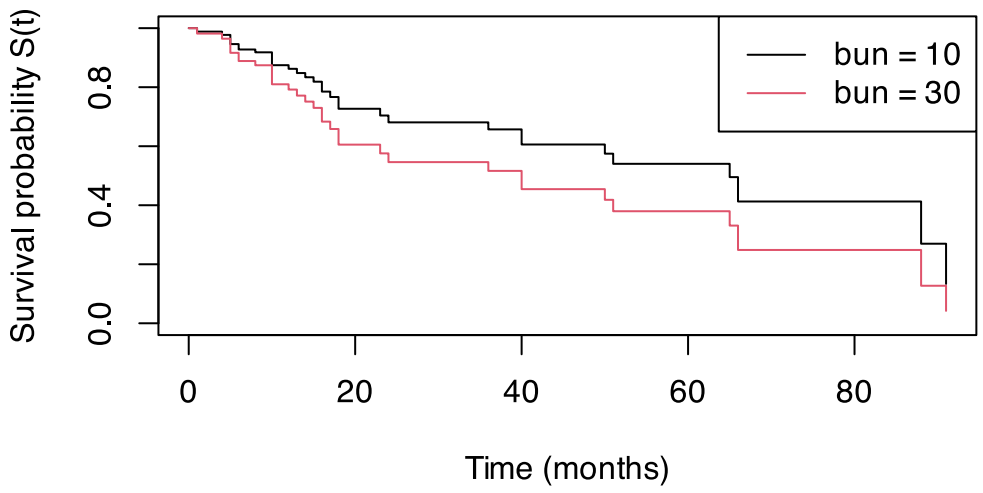
```
7 x 1 sparse Matrix of class "dgCMatrix"
                  1
age      .
sex      .
bun      0.01568385
ca       .
hb      -0.08388993
pcells   .
protein -0.37897248
```

Interpretation. The left edge of the plot is the fully penalised (null) model; as $\log \lambda$ decreases, more covariates enter. At `lambda.min` the coefficients above are shown (a `.` means the covariate was set to zero and excluded): the lasso retains `bun`, `hb` and `protein` and drops the rest. A 10-fold cross-validated choice of λ guards against overfitting. Note that lasso coefficients are biased toward zero, so standard errors and p-values are usually not reported after selection.

Predicted survival curves from the model

Given a fitted Cox model, `survival::survfit()` returns estimated survival curves for specified covariate patterns (eq. 3.26). Here we compare patients with low and high blood urea nitrogen, holding the other covariates fixed:

```r
plot(
  survival::survfit(
    fit,
    newdata = data.frame(
      age = 60,
      sex = 1,
      bun = c(10, 30),
      ca = 10,
      hb = 12,
      pcells = 50,
      protein = 1
    )
  ),
  col = 1:2,
  xlab = "Time (months)",
  ylab = "Survival probability S(t)"
)
legend("topright", legend = c("bun = 10", "bun = 30"), col = 1:2, lty = 1)
```



Interpretation. The curves use $\hat{S}_i(t) = \hat{S}_0(t)^{\exp(\hat{\eta}_i)}$ (eq. 3.26): the patient with higher blood urea nitrogen has a lower estimated survival curve, all else equal. The two curves never cross because of the proportional-hazards structure.

An alternative framework: `rms`

Frank Harrell's `rms` package offers `rms::cph()` (Cox) with a consistent interface for validation, calibration and plotting:

```
dd <- rms::datadist(myeloma)
options(datadist = "dd")
f <- rms::cph(
  Surv(time, status) ~ age + sex + bun + ca + hb + pcells + protein,
  data = myeloma,
  x = TRUE,
  y = TRUE
)
f
```

```
Cox Proportional Hazards Model

rms::cph(formula = Surv(time, status) ~ age + sex + bun + ca +
    hb + pcells + protein, data = myeloma, x = TRUE, y = TRUE)

                      Model Tests    Discrimination
                                            Indexes
Obs        48    LR chi2     17.53    R2       0.309
Events     36    d.f.            7    R2(7,48) 0.197
Center -2.219    Pr(> chi2) 0.0143    R2(7,36) 0.254
                 Score chi2  25.59    Dxy      0.410
                 Pr(> chi2) 0.0006

        Coef    S.E.   Wald Z Pr(>|Z|)
age     -0.0181 0.0278 -0.65  0.5165
sex     -0.2495 0.4031 -0.62  0.5360
bun      0.0227 0.0061  3.71  0.0002
ca       0.0133 0.1327  0.10  0.9204
hb      -0.1330 0.0685 -1.94  0.0523
pcells  -0.0014 0.0066 -0.21  0.8366
protein -0.6833 0.4294 -1.59  0.1116
```

Interpretation. `rms::cph()` reports the same hazard ratios as `survival::coxph()` but stores the design matrix, enabling `rms::validate()` (bootstrap optimism correction) and `rms::calibrate()` (calibration plots), which are the package's main advantages for prediction modelling.

Chapter 4 — Model checking in the Cox regression model

Residuals, influential observations and the proportional hazards assumption (§4.1–4.5).

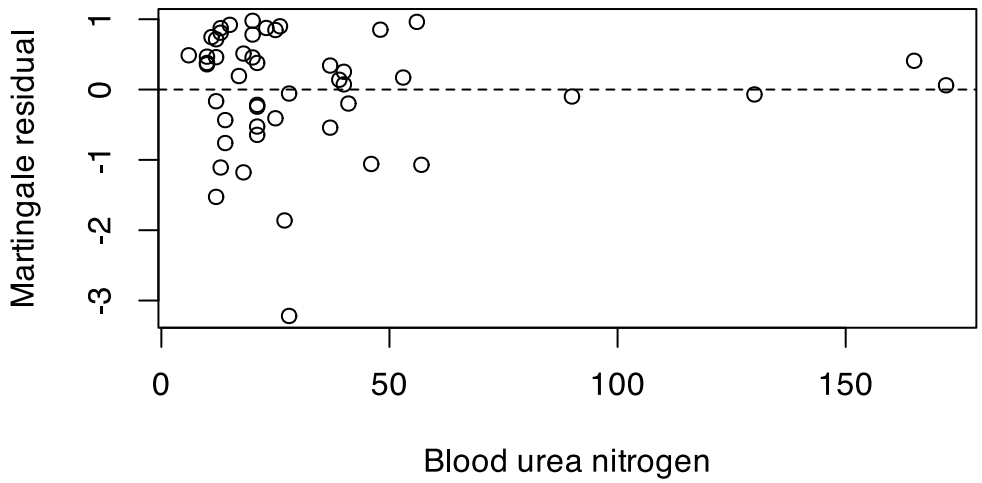
Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Cox–Snell / martingale / deviance residuals	`residuals(..., type = "...")`	`rms::resid()`
Schoenfeld residuals	`residuals(..., type = "schoenfeld")`	`survival::cox.zph()`
Influence (dfbeta)	`residuals(..., type = "dfbeta")`	`rms::which.influence()`

Residuals and influence

We reuse the myeloma model `fit` from Chapter 3. Martingale residuals (eq. 4.6) check the functional form of a covariate; dfbeta measures each observation's influence:

```
r_mart <- residuals(fit, type = "martingale")
r_dev  <- residuals(fit, type = "deviance")
plot(myeloma$bun, r_mart, xlab = "Blood urea nitrogen", ylab = "Martingale
residual")
abline(h = 0, lty = 2)
```




```
dfb <- residuals(fit, type = "dfbeta")
round(head(dfb), 3)
```

```
    [,1]   [,2] [,3]   [,4]   [,5]   [,6]   [,7]
1  0.003 -0.022    0 -0.011  0.012  0.000  0.085
2 -0.005 -0.014    0 -0.005 -0.013 -0.002  0.021
3 -0.006  0.119    0  0.047  0.001  0.000  0.106
4  0.002 -0.006    0 -0.006  0.001  0.000  0.037
5  0.002 -0.019    0 -0.008  0.013  0.001 -0.025
6  0.000 -0.011    0  0.005 -0.001  0.000  0.002
```

Interpretation. Martingale residuals hovering around zero with no systematic trend support a linear effect of `bun`. Each row of `dfb` is the change in a coefficient when that patient is removed; large absolute values flag influential observations. Deviance residuals (`r_dev`, eq. 4.7) are centred and more symmetric, making outliers easier to spot.

Chapter 5 — Parametric regression models

Exponential, Weibull and related distributions; PH and AFT models (§5.1–5.17).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Parametric PH/AFT models	`survival::survreg()`	`flexsurv::flexsurvreg()`
Flexible spline models	—	`flexsurv::flexsurvspline()` (Chapter 6)

Fitting exponential and Weibull models

For the IUD data we fit exponential and Weibull models and compare them:

```
exp_fit <- survival::survreg(Surv(time, status) ~ 1, data = iud, dist =
"exponential")
wei_fit <- survival::survreg(Surv(time, status) ~ 1, data = iud, dist = "weibull")
summary(wei_fit)
```

```

Call:
survival::survreg(formula = Surv(time, status) ~ 1, data = iud,
    dist = "weibull")
             Value Std. Error     z      p
(Intercept)  4.592      0.204 22.47 <2e-16
Log(scale)  -0.517      0.275 -1.88   0.06

Scale= 0.597

Weibull distribution
```

```
Loglik(model)= -50.4   Loglik(intercept only)= -50.4
Number of Newton-Raphson Iterations: 6
n= 18
```

```
AIC(exp_fit, wei_fit)
```

```
        df      AIC
exp_fit  1 105.5991
wei_fit  2 104.7046
```

`flexsurv::flexsurvreg()` reports the same model on the hazard scale:

```
flexsurv::flexsurvreg(Surv(time, status) ~ 1, data = iud, dist = "weibull")
```

```
Call:
flexsurv::flexsurvreg(formula = Surv(time, status) ~ 1, data = iud,
    dist = "weibull")

Estimates:
       est      L95%     U95%     se
shape    1.676    0.979    2.872    0.460
scale   98.644   66.086  147.242   20.160

N = 18,  Events: 9,  Censored: 9
Total time at risk: 1046
Log-likelihood = -50.35232, df = 2
AIC = 104.7046
```

Interpretation. `survreg` is parameterised as an accelerated failure time model, $\log T = \mu + \sigma W$ (eq. 5.61); the Weibull shape is $1/\text{scale}$. The smaller AIC identifies the preferred model. `flexsurvreg` uses the hazard-scale parameterisation (shape γ and rate λ, eq. 5.4), which is closer to the book's notation.

Chapter 6 — Flexible parametric models

Piecewise and spline-based models (§6.1–6.5).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Royston–Parmar spline models	—	`flexsurv::flexsurvspline()`, `rstpm2::stpm2()`

Task	Collett (Ch. 17)	Modern alternative
Piecewise exponential	`survival::survreg()` (in intervals)	`flexsurv::flexsurvspline()`

A Royston–Parmar model

We model the log cumulative hazard of the hypernephroma patients with a restricted cubic spline in log time:

```
flexsurv::flexsurvspline(Surv(time, status) ~ nephrectomy, data = hyper, k = 1)
```

```
Call:
flexsurv::flexsurvspline(formula = Surv(time, status) ~ nephrectomy,
    data = hyper, k = 1)

Estimates:
             data mean  est       L95%      U95%      se        exp(est)
gamma0             NA   -5.13483  -7.65126  -2.61839   1.28392        NA
gamma1             NA    2.03033   1.08910   2.97156   0.48023        NA
gamma2             NA    0.16092  -0.00861   0.33045   0.08650        NA
nephrectomy   0.80556   -1.75519  -2.74702  -0.76336   0.50604   0.17287
             L95%      U95%
gamma0             NA        NA
gamma1             NA        NA
gamma2             NA        NA
nephrectomy   0.06412   0.46610

N = 36,  Events: 32,  Censored: 4
Total time at risk: 1340
Log-likelihood = -144.2456, df = 4
AIC = 296.4911
```

Interpretation. The spline terms give a smooth baseline log cumulative hazard (eq. 6.13), avoiding the rigid shape assumptions of a single parametric distribution. `nephrectomy` enters as a proportional-hazards term, so its exponentiated coefficient is the hazard ratio for nephrectomy.

Chapter 7 — Model checking in parametric models

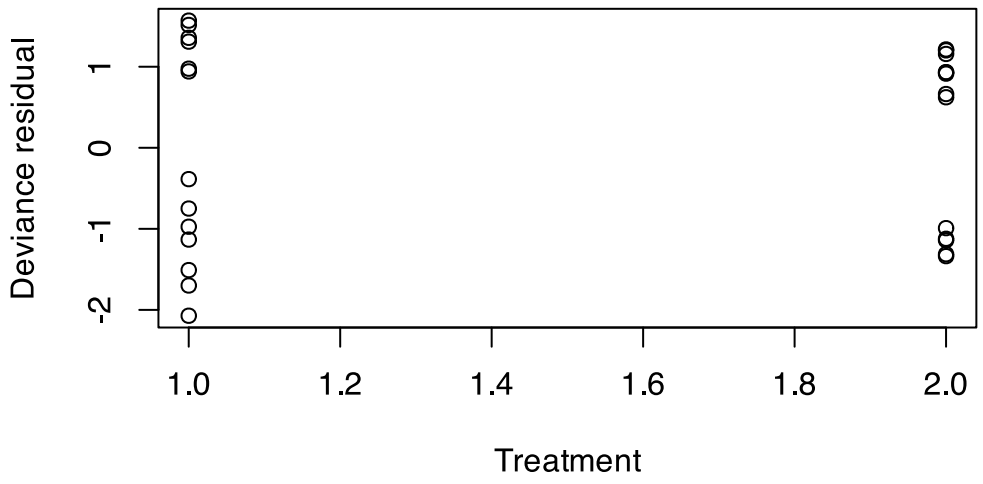
Residuals for parametric models (§7.1–7.5).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Standardised / Cox–Snell / martingale / deviance residuals	`residuals(survival::survreg(...), type = "...")`	

Residuals for a Weibull model

```
ov_fit <- survival::survreg(Surv(time, status) ~ treat, data = ovarian, dist
= "weibull")
r_dev_ov <- residuals(ov_fit, type = "deviance")
plot(ovarian$treat, r_dev_ov, xlab = "Treatment", ylab = "Deviance residual")
```



Interpretation. Deviance residuals (eq. 7.5) should be roughly symmetric around zero with no pattern across covariates; outliers or a trend indicate a poor fit.

Chapter 8 — Time-dependent variables

Covariates and coefficients that change over time (§8.1–8.7).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Time-dependent covariates	`survival::Surv(start, stop, event) + survival::coxph()`	`JM`
Time-varying coefficients	`survival::coxph(..., tt = ...)`	—

A time-varying coefficient

The bone-marrow transplant data (`bmt`) records recipient age `rage`. We allow its effect to vary with log time via `tt()`:

```
m_tt <- survival::coxph(Surv(time, status) ~ rage + tt(rage), data = bmt,
                        tt = function(x, t, ...) x * log(t))
summary(m_tt)
```

```
Call:
survival::coxph(formula = Surv(time, status) ~ rage + tt(rage),
    data = bmt, tt = function(x, t, ...) x * log(t))

  n= 37, number of events= 17

             coef exp(coef) se(coef)      z Pr(>|z|)
rage      0.23255   1.26182  0.20442  1.138    0.255
tt(rage) -0.03412   0.96646  0.04029 -0.847    0.397

         exp(coef) exp(-coef) lower .95 upper .95
rage        1.2618     0.7925    0.8453     1.884
tt(rage)    0.9665     1.0347    0.8931     1.046

Concordance= 0.599  (se = 0.087 )
Likelihood ratio test= 3.71  on 2 df,   p=0.2
Wald test            = 3.99  on 2 df,   p=0.1
Score (logrank) test = 4.19  on 2 df,   p=0.1
```

Interpretation. `rage` is the constant part of the effect and `tt(rage)` its interaction with $\log t$ (eq. 8.1). Here `tt(rage)` is not significant, so there is little evidence that the effect of recipient age changes over time. Genuinely time-dependent covariates (e.g. a treatment that starts later) are coded in counting-process format `Surv(start, stop, event)` with one row per time interval.

Chapter 9 — Interval-censored survival data

Turnbull NPMLE and semi-parametric models (§9.1–9.4).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Non-parametric MLE (Turnbull)	`interval::icfit()`	`icenReg::ic_np()`
Semi-parametric (PH) models	`interval::...`	`icenReg::ic_sp()`

The Turnbull estimate

We simulate interval-censored data (assessments every 6 months) and estimate $S(t)$:

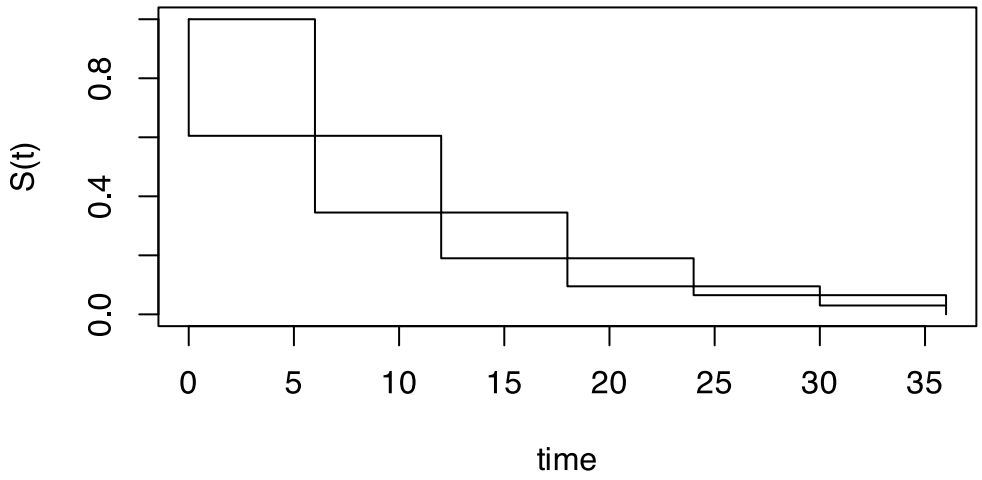
```
set.seed(5)
true_t <- rexp(200, rate = 0.1)
grid   <- seq(0, 36, by = 6)
L <- vapply(true_t, function(t) max(grid[grid <= t]), numeric(1))
R <- vapply(true_t, function(t) min(grid[grid >  t]), numeric(1))
R[is.infinite(R)] <- Inf
R[true_t > 36] <- Inf
L[true_t > 36] <- 36
fit_np <- icenReg::ic_np(cbind(L, R))
fit_np
```

```

Model:  Non-parametric
Dependency structure assumed: Independence
No covariates used

final llk =  -311.492
Iterations =  2
```

```
plot(fit_np)
```



Interpretation. Each event time is known only up to an interval $(L, R]$. The Turnbull NPMLE (eq. 9.2) puts mass only on intervals consistent with the data; the plot shows the estimated survivor function and its pointwise confidence limits.

Chapter 10 — Frailty models

Unobserved heterogeneity and clustered data (§10.1–10.8).

Functions and packages

Task		Collett (Ch. 17)	Modern alternative
Parametric frailty		`parfm::parfm()`	`survival::...`
Semi-parametric frailty	(shared)	`survival::coxph(..., frailty(...))`	`coxme::coxme()`

Shared frailty for paired kidneys

Kidneys from the same donor are correlated; `frailty(donor)` introduces a shared random effect (eq. 10.31):

```
kfit <- survival::coxph(Surv(time, status) ~ age + diabetes + frailty(donor),
data = kidney)
summary(kfit)
```

```
Call:
survival::coxph(formula = Surv(time, status) ~ age + diabetes +
    frailty(donor), data = kidney)

  n= 434, number of events= 71

                coef     se(coef) se2     Chisq DF    p
age              0.01901 0.01039  0.01007  3.35  1.00 0.067
diabetes        -0.16428 0.44067  0.42867  0.14  1.00 0.710
frailty(donor)                            23.33 22.31 0.400

          exp(coef) exp(-coef) lower .95 upper .95
age          1.0192     0.9812    0.9986     1.040
diabetes     0.8485     1.1785    0.3577     2.013

Iterations: 6 outer, 22 Newton-Raphson
     Variance of random effect= 0.3396451   I-likelihood = -416.3
Degrees of freedom for terms=  0.9  0.9 22.3
Concordance= 0.833  (se = 0.026 )
Likelihood ratio test= 44.86  on 24.19 df,   p=0.006
```

Interpretation. The `frailty(donor)` term captures correlation between recipients of kidneys from the same donor; its estimated coefficient is the frailty variance (larger values mean stronger within-donor correlation). The fixed effects `age` and `diabetes` are interpreted as usual.

Chapter 11 — Non-proportional hazards and institutional comparisons

Stratification, restricted mean survival, and pseudo-values (§11.1–11.4).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Stratified Cox	`survival::coxph(..., strata(...))`	—
Restricted mean survival	`pseudo::pseudomean()`	`RISCA::rmst()`

Non-proportional hazards and the restricted mean

The gastric-cancer treatments have crossing hazards. We check proportional hazards and then compare restricted mean survival times via pseudo-values:

```
cox.zph(survival::coxph(Surv(time, status) ~ treat, data = gastric))
```

```
       chisq df       p
treat   13.2  1 0.00028
GLOBAL  13.2  1 0.00028
```

```
pm <- pseudo::pseudomean(time = gastric$time, event = gastric$status, tmax = 400)
summary(glm(pm ~ gastric$treat))
```

```

Call:
glm(formula = pm ~ gastric$treat)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)     345.51      18.19  18.995  < 2e-16 ***
gastric$treat   -83.91      25.72  -3.262  0.00158 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 14888.98)

    Null deviance: 1468654  on 89  degrees of freedom
Residual deviance: 1310230  on 88  degrees of freedom
AIC: 1124.1

Number of Fisher Scoring iterations: 2
```

Interpretation. The `cox.zph` output shows evidence against proportional hazards, so a single hazard ratio is a time-average. The restricted mean survival time over $[0, 400]$ days (eq. 11.3) is then a more interpretable summary; the `glm` on pseudo-values compares the two treatment means directly (its `treat` coefficient is the difference in RMST).

Chapter 12 — Competing risks

Cause-specific hazards, cumulative incidence, and Fine–Gray (§12.1–12.6).

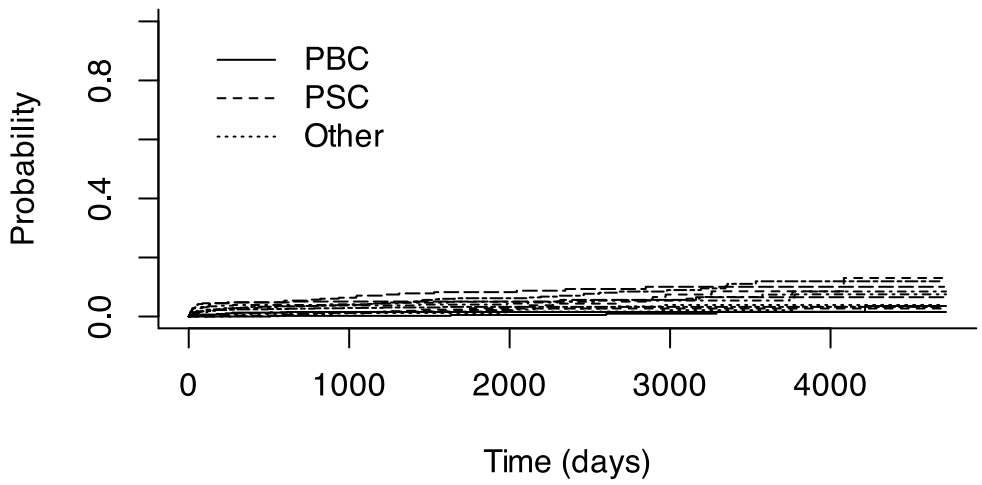
Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Cumulative incidence	`cmprsk::cuminc()`	`tidycmprsk::cuminc()`, `survival::survfit()`
Cause-specific / Fine–Gray model	`cmprsk::crr()`	`tidycmprsk::crr()`, `riskRegression::FGR()`

Cumulative incidence of graft-failure causes

`liver` records the cause of graft failure in `cof` (0 = censored, 1–4 = causes). We estimate cumulative incidence functions by indication for transplant (`disease`):

```
ci <- cmprsk::cuminc(liver$time, liver$cof, liver$disease)
plot(ci, curvlab = c("PBC", "PSC", "Other"), xlab = "Time (days)")
```



A Fine–Gray model for one cause (eq. 12.11):

```
cmprsk::crr(liver$time, liver$cof, cov1 = as.matrix(liver[c("gender", "age")]))
```

```
convergence:  TRUE
coefficients:
  gender      age
 0.33260 -0.03897
standard errors:
[1] 0.37250 0.01878
two-sided p-values:
gender    age
 0.370  0.038
```

Interpretation. Each curve is the probability of graft failure from a given cause by time t (the cumulative incidence function, eq. 12.4); the cause-specific curves add up to the total failure

probability. The Fine–Gray model relates covariates to the subdistribution hazard of a specific cause.

Chapter 13 — Multiple events and event history modelling

Counting processes and recurrent events (§13.1–13.5).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Recurrent events (AG, PWP)	`survival::coxph(..., cluster(...))`	`mstate::...`, `frailtypack::...`

The Anderson–Gill model

`survival::bladder` holds recurrent tumour episodes in counting-process form. The Anderson–Gill model treats each recurrence as an event with a common baseline hazard and a robust (cluster) variance:

```
ag <- survival::coxph(Surv(stop, event) ~ rx + size + number + cluster(id), data
= bladder)
summary(ag)
```

```
Call:
survival::coxph(formula = Surv(stop, event) ~ rx + size + number,
    data = bladder, cluster = id)

  n= 340, number of events= 112

           coef exp(coef) se(coef) robust se      z Pr(>|z|)
rx     -0.53983   0.58285  0.19979   0.29730 -1.816  0.06940 .
size   -0.05487   0.94660  0.06961   0.08824 -0.622  0.53402
number  0.19349   1.21347  0.04638   0.06747  2.868  0.00414 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

       exp(coef) exp(-coef) lower .95 upper .95
rx        0.5828     1.7157    0.3255     1.044
size      0.9466     1.0564    0.7963     1.125
number    1.2135     0.8241    1.0632     1.385

Concordance= 0.642  (se = 0.034 )
Likelihood ratio test= 22.32  on 3 df,   p=6e-05
Wald test            = 12.51  on 3 df,   p=0.006
Score (logrank) test = 25.18  on 3 df,   p=1e-05,   Robust = 11.01  p=0.01
```

```
  (Note: the likelihood ratio and score tests assume independence of
     observations within a cluster, the Wald and robust score tests do not).
```

```
hr_rx  <- exp(coef(ag))[["rx"]]
hr_num <- exp(coef(ag))[["number"]]
```

Interpretation. `rx` is the treatment effect (hazard ratio 0.58) and `number` the effect of prior recurrences (hazard ratio 1.21 per extra tumour). `cluster(id)` gives robust standard errors that account for within-patient correlation (eq. 13.5). The same study appears in the book's folder as "Recurrence of bladder cancer.dat" in a time-to-first-recurrence form.

Chapter 14 — Dependent censoring

Sensitivity analysis and inverse probability weighting (§14.1–14.3).

Functions and packages

Task	Collett (Ch. 17)		Modern alternative
Inverse probability of censoring weighting	`survival::coxph()` `weights`	+	—

Inverse probability of censoring weighting

For patients on the liver-transplant waiting list, censoring (transplant) is likely dependent on the risk of death. We model the censoring process, form inverse probability weights, and refit the model for death:

```
liver_wait$cens <- 1 - liver_wait$status   # censoring = transplant / removal
cf <- survival::coxph(Surv(time, cens) ~ age + gender + bmi + ukeld, data
= liver_wait)
sc <- survival::survfit(cf, newdata = liver_wait)
idx <- sapply(seq_len(nrow(liver_wait)), function(i) max(which(sc$time <=
liver_wait$time[i])))
w   <- 1 / sc$surv[idx]
summary(w)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.004   1.267   1.879   7.624   4.010 228.130
```

```
wfit <- survival::coxph(Surv(time, status) ~ age + gender + bmi + ukeld,
                        data = liver_wait, weights = w)
summary(wfit)
```

```
Call:
survival::coxph(formula = Surv(time, status) ~ age + gender +
    bmi + ukeld, data = liver_wait, weights = w)

  n= 281, number of events= 64

            coef exp(coef)  se(coef) robust se      z Pr(>|z|)
age    -0.048531  0.952628  0.006368  0.047524 -1.021   0.3072
gender -1.110729  0.329319  0.357522  1.006660 -1.103   0.2699
bmi    -0.022709  0.977547  0.010762  0.106844 -0.213   0.8317
ukeld   0.163719  1.177884  0.016494  0.079213  2.067   0.0387 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

       exp(coef) exp(-coef) lower .95 upper .95
age       0.9526      1.050   0.86790     1.046
gender    0.3293      3.037   0.04579     2.369
bmi       0.9775      1.023   0.79285     1.205
ukeld     1.1779      0.849   1.00850     1.376

Concordance= 0.637  (se = 0.092 )
Likelihood ratio test= 177.2  on 4 df,   p=<2e-16
Wald test            = 18.42  on 4 df,   p=0.001
Score (logrank) test = 235  on 4 df,   p=<2e-16,   Robust = 3.07  p=0.5

  (Note: the likelihood ratio and score tests assume independence of
     observations within a cluster, the Wald and robust score tests do not).
```

Interpretation. Each weight is the inverse of the estimated probability of remaining uncensored (eq. 14.3). The very large weights seen in the summary belong to patients whose censoring probability was small, and are a warning that dependent censoring may be influencing the estimates — the same point made by the book's sensitivity analysis.

Chapter 15 — Sample size requirements for a survival study

Number of events and number of patients (§15.1–15.3).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Number of events (Schoenfeld)	—	`powerSurvEpi::ssizeEpi()`, `gsDesign::...`

Required number of deaths

Schoenfeld's formula (eq. 15.1) is a one-liner in base R. For the chronic-active-hepatitis example, detecting a hazard ratio of 0.57 with 80% power at the two-sided 5% level:

```
alpha <- 0.05; power <- 0.8; hr <- 0.57; p1 <- p2 <- 0.5
d_deaths <- (qnorm(1 - alpha / 2) + qnorm(power))^2 / (p1 * p2 * log(hr)^2)
round(d_deaths, 1)
```

```
[1] 99.4
```

Interpretation. About 99 deaths are required; the number of patients follows by dividing by the expected probability of death during follow-up (eq. 15.10). The CRAN package `powerSurvEpi` provides the same calculations with more options.

Chapter 16 — Bayesian survival analysis

Priors, posteriors, and MCMC (§16.1–16.10).

Functions and packages

Task	Collett (Ch. 17)	Modern alternative
Bayesian parametric models (MCMC)	`rstanarm::stan_surv()`	`survstan::...`, `INLA`
Conjugate analysis (exponential)	base R (closed form)	—

A conjugate analysis of the exponential model

For the IUD data under an exponential model with rate λ and a vague gamma prior, the posterior is gamma with shape r and rate $\sum t_i$ (the closed form of eq. 16.4):

```
r     <- sum(iud$status)
total <- sum(iud$time)
a0 <- b0 <- 0.001                    # vague Gamma(a0, b0) prior
a1 <- a0 + r; b1 <- b0 + total
post_mean <- a1 / b1
post_ci   <- qgamma(c(0.025, 0.975), a1, b1)
c(mean = post_mean, lower = post_ci[1], upper = post_ci[2])
```

```
       mean       lower       upper
0.008605154 0.003935029 0.015071228
```

Interpretation. The posterior mean of λ is about 0.0086 per week (95% credible interval 0.0039–0.0151), matching the maximum-likelihood estimate. For non-conjugate models, `rstanarm::stan_surv()` fits a Bayesian version via MCMC.

Chapter 17 (Survival analysis with R) is the source of the function–package mappings used throughout; it is not repeated here.